EXISTENCE OF TRANSCENDENCE BASES AND EQUALITY OF THEIR CARDINALITIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The exchange step reuses the irreducible polynomial f: if g(beta_1, alpha_2, ..., alpha_n) = 0 with g in F[x, y_2, ..., y_n] involved x, then beta_1 would be algebraic over F(alpha_2, ..., alpha_n), contradicting the choice of f involving y_1.

Editorial note. The label lemma-transcendence-degree names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 3585-3605: a transcendence basis of K/k is an algebraically independent collection {x_i} such that K/k(x_i; i in I) is algebraic
- Zorn's lemma applied to the collection of algebraically independent subsets of G containing A; the union of a totally ordered family stays algebraically independent because polynomials involve only finitely many variables

Definition 0.1. Let K/k be a field extension.

(1) A collection of elements $\{x_i\}_{i \in I}$ of K is called *algebraically independent* over k if the map
$$k[X_i; i \in I] \longrightarrow K$$
which maps X_i to x_i is injective.
(2) The field of fractions of a polynomial ring $k[x_i; i \in I]$ is denoted $k(x_i; i \in I)$.
(3) A *purely transcendental extension* of k is any field extension K/k isomorphic to the field of fractions of a polynomial ring over k.
(4) A *transcendence basis* of K/k is a collection of elements $\{x_i\}_{i \in I}$ which are algebraically independent over k and such that the extension $K/k(x_i; i \in I)$ is algebraic.

Lemma 0.2. *Let E/F be a field extension. A transcendence basis of E over F exists. Any two transcendence bases have the same cardinality.*

Proof. Let A be an algebraically independent subset of E. Let G be a subset of E containing A that generates E/F. We claim we can find a transcendence basis B such that $A \subset B \subset G$. To prove this, consider the collection $\mathcal{B}$ of algebraically independent subsets whose members are subsets of G that contain A. Define a partial ordering on $\mathcal{B}$ using inclusion. Then $\mathcal{B}$ contains at least one element A. The union of the elements of a totally ordered subset T of $\mathcal{B}$ is an algebraically independent subset of E over F since any algebraic dependence relation would

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have occurred in one of the elements of T (since polynomials only involve finitely many variables). The union also contains A and is contained in G. By Zorn's lemma, there is a maximal element $B \in \mathcal{B}$. Now we claim E is algebraic over $F(B)$. This is because if it wasn't then there would be an element $f \in G$ transcendental over $F(B)$ since $F(G) = E$. Then $B \cup \{f\}$ would be algebraically independent contradicting the maximality of B. Thus B is our transcendence basis.

Let B and B' be two transcendence bases. Without loss of generality, we can assume that $|B'| \leq |B|$. Now we divide the proof into two cases: the first case is that B is an infinite set. Then for each $\alpha \in B'$, there is a finite set $B_\alpha \subset B$ such that α is algebraic over $F(B_\alpha)$ since any algebraic dependence relation only uses finitely many indeterminates. Then we define $B^* = \bigcup_{\alpha \in B'} B_\alpha$. By construction, $B^* \subset B$, but we claim that in fact the two sets are equal. To see this, suppose that they are not equal, say there is an element $\beta \in B \setminus B^*$. We know β is algebraic over $F(B')$ which is algebraic over $F(B^*)$. Therefore β is algebraic over $F(B^*)$, a contradiction. So $|B| \leq |\bigcup_{\alpha \in B'} B_\alpha|$. Now if B' is finite, then so is B so we can assume B' is infinite; this means

$$|B| \leq |\bigcup\nolimits_{\alpha \in B'} B_\alpha| = |B'|$$

because each B_α is finite and B' is infinite. Therefore in the infinite case, $|B| = |B'|$.

Now we need to look at the case where B is finite. In this case, B' is also finite, so suppose $B = \{\alpha_1, \ldots, \alpha_n\}$ and $B' = \{\beta_1, \ldots, \beta_m\}$ with $m \leq n$. We perform induction on m: if $m = 0$ then E/F is algebraic so $B = \emptyset$ so $n = 0$. If $m > 0$, there is an irreducible polynomial $f \in F[x, y_1, \ldots, y_n]$ such that $f(\beta_1, \alpha_1, \ldots, \alpha_n) = 0$ and such that x occurs in f. Since β_1 is not algebraic over F, f must involve some y_i so without loss of generality, assume f uses y_1. Let $B^* = \{\beta_1, \alpha_2, \ldots, \alpha_n\}$. We claim that B^* is a basis for E/F. To prove this claim, we see that we have a tower of algebraic extensions

$$E/F(B^*, \alpha_1)/F(B^*)$$

since α_1 is algebraic over $F(B^*)$. Now we claim that B^* (counting multiplicity of elements) is algebraically independent over F because if it weren't, then there would be an irreducible $g \in F[x, y_2, \ldots, y_n]$ such that $g(\beta_1, \alpha_2, \ldots, \alpha_n) = 0$ which must involve x making β_1 algebraic over $F(\alpha_2, \ldots, \alpha_n)$ which would make α_1 algebraic over $F(\alpha_2, \ldots, \alpha_n)$ which is impossible. So this means that $\{\alpha_2, \ldots, \alpha_n\}$ and $\{\beta_2, \ldots, \beta_m\}$ are bases for E over $F(\beta_1)$ which means by induction, $m = n$. $\square$